

Geographic Equity Rotation

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<https://github.com/GerardWu100/geographic-equity-rotation>

This project examines whether foreign-exchange option skew can help predict the relative performance of regional equity markets. The strategy rotates among SPY, representing the United States; EZU, representing the Eurozone; and EWJ, representing Japan. The main hypothesis is that foreign-exchange risk reversals reflect directional demand for currencies and may therefore contain useful information about future regional equity returns. A one-month, 25-delta risk reversal is calculated as the implied volatility of a call option minus the implied volatility of a put option. The signs of the currency signals are adjusted so that a positive value consistently represents stronger demand for the local currency of each equity region.

Data

The analysis combines two Bloomberg data files covering February 27, 2012 to July 28, 2026. The first contains closing prices and trading volumes for SPY, EZU, and EWJ, together with the VIX, V2X, and Japanese equity-volatility indices. The second contains one-month, 25-delta risk reversals for EURUSD, USDJPY, and EURJPY. After merging the files by date, the final dataset contains 3,625 valid market sessions.

The data-cleaning process removed 137 rows representing weekdays on which the New York Stock Exchange was closed. On these dates, the prices and volumes of all three exchange-traded funds were exact copies of the previous row. Keeping these observations would have created artificial zero-return sessions and distorted momentum, volatility, and rebalance timing. The Japanese volatility series, NKYVF, was also excluded because it contained an unexplained increase of approximately 100 times its previous level. Rather than manually repair an unverified series, the final model omitted it.

Methodology

A separate feature set is constructed for each exchange-traded fund. The equity features include the previous 20-session return, the previous 60-session return, price relative to its 50-session moving average, and volume relative to its 20-session moving average. Each model also receives two sign-adjusted foreign-exchange risk-reversal signals. SPY and EZU additionally use the level, 20-session change, and 60-session standardized value of their regional volatility indices. The final models therefore use nine features for SPY, nine for EZU, and six for EWJ.

The model observes all features at the close of a decision date, enters the position at the next valid close, and measures the return over the following 20 sessions. This timing avoids assuming that the strategy can observe closing information and trade at the same closing price. A 21-

session gap is placed between training and test samples so that forward-return labels from the training period cannot overlap with the test period.

A separate Ridge regression model is fitted for each fund. Ridge regression is a linear model that shrinks unstable coefficients toward zero, making it suitable for correlated features and noisy financial data. The model is evaluated using five expanding walk-forward test periods. Within each training period, four-fold time-series cross-validation selects the Ridge penalty from 30 possible values. Feature standardization, model selection, and coefficient estimation are repeated inside each fold to reduce information leakage.

Predictive Results

Predictive performance is measured using the Spearman rank information coefficient. In the final implementation, this statistic is calculated separately for each fund across time rather than directly across the three funds on each date. The average out-of-sample information coefficients were 0.196 for SPY, 0.182 for EZU, and 0.081 for EWJ. SPY and EZU produced positive results in every test fold, while EWJ was weaker and less stable. These findings suggest that the models contain modest predictive information, particularly for the United States and Eurozone, although the results do not prove that forecasts from the three separate models are perfectly comparable for cross-regional ranking.

The portfolio is rebalanced every 20 valid prediction dates. At each rebalance, the three funds are ranked by predicted return and the two highest-ranked funds receive initial weights of 50% each. Portfolio risk is estimated using the previous 63 daily returns, and exposure is reduced when estimated annualized volatility exceeds 12%. Leverage is not permitted, and any unallocated capital is held in cash with a zero assumed return. Transaction costs of five basis points are charged for exchange-traded fund purchases and sales.

Portfolio Results

The risk-managed top-two strategy produced a compound annual growth rate of 7.75%, annualized volatility of 11.96%, and a Sharpe ratio of 0.689. The same top-two strategy without volatility scaling produced a higher compound annual growth rate of 11.65% and a higher Sharpe ratio of 0.819, although its volatility increased to 14.91%. The equal-weight portfolio returned 10.12% annually, while the single-asset rotation strategy returned 12.51%.

SPY was the strongest benchmark, producing a compound annual growth rate of 13.94% and a Sharpe ratio of 0.985. The volatility-targeting process successfully lowered the strategy's volatility, but it did not materially improve maximum drawdown and reduced overall risk-adjusted performance. The risk-managed portfolio held an average cash allocation of approximately 17.7%, which created a significant return drag.

The strategy was also tested using all 20 possible rebalance starting points. Across these tests, compound annual growth rates ranged from 7.07% to 8.66%, while Sharpe ratios ranged from 0.596 to 0.805. This indicates that the result was not driven entirely by one favourable rebalance

date, although it remains a historical robustness test rather than a genuinely independent future test.

Limitations

The results should be viewed as exploratory. Earlier notebooks used the same historical sample to examine features and select the 20-session horizon, meaning that the full dataset is not an untouched research sample. The Japanese model also contains fewer features because its volatility series was excluded. In addition, the strategy covers only three regional funds, so the results may not generalize to a broader international equity universe.

The backtest assumes next-close execution, fixed transaction costs, no market impact, and a zero return on cash. The dividend and adjustment treatment of the supplied Bloomberg closing-price fields also requires confirmation. Furthermore, the reported drawdowns are based on returns recorded at the end of each 20-session holding period rather than on a daily marked-to-market portfolio, so losses within a holding period could have been larger. Finally, the information coefficient evaluates each fund separately across time, while the portfolio compares forecasts produced by three different models. Further calibration tests would be required to confirm that these predictions are directly comparable.

Conclusion

The project finds that foreign-exchange option skew, combined with momentum, trend, relative volume, and volatility features, contains modest information about subsequent regional equity returns. The evidence is strongest for SPY and EZU and weaker for EWJ. The final pipeline applies reasonable controls for data quality, execution timing, overlapping labels, model validation, and transaction costs.

However, the predictive evidence did not translate into a benchmark-beating portfolio. The risk-managed strategy reduced volatility but underperformed SPY in both return and Sharpe ratio, while its cash allocation created a meaningful performance drag. The appropriate conclusion is that currency skew may be useful as one input within a broader asset-allocation framework, but the current evidence is not strong enough to support it as a standalone regional equity-rotation strategy.